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Simultaneously exfoliated boron doped graphene sheets to encapsulate sulfur for applications in lithium sulfur batteries

Pengcheng Shi[†], Yong Wang[†], Xin Liang[†], Yi Sun[†], Sheng Cheng[†], Chunhua Chen[‡],
Hongfa Xiang^{†*}

[†]School of Materials Science and Engineering, Hefei University of Technology, No. 193, Tunxi Road, Hefei, Anhui 230009, China

[‡]CAS Key Laboratory of Materials for Energy Conversions, Department of Materials Science and Engineering, University of Science and Technology of China, No. 96, Jinzhai Road, Hefei, Anhui 230026, China

*Corresponding author: Tel.: +86-551-62901457; E-mail: hfxiang@hfut.edu.cn

Abstract

Heteroatom-doped high-quality graphene is highly effective in encapsulating sulfur (S) and fabricating a high-performance electrode for lithium-sulfur (Li-S) batteries. Herein, simultaneously exfoliated boron doped graphene sheets (B-EEG) is prepared via electrochemical exfoliation of graphite in 1.0 mol L⁻¹ Li bis(oxalato)borate (LiBOB)/dimethyl methylphosphonate (DMMP) electrolyte. The obtained B-EEG possesses high quality with large planar size of ~11 μm, few structure defects of I_D/I_G=0.26 and low oxygen content of 7.93%. After B-EEG is used to encapsulate S through an in-situ deposition route, the S@B-EEG with an S content of 72.5% displays a high initial discharge capacity of 1476 mAh g⁻¹ at 0.1 C. Compared to conventional S@thermally reduced graphene oxide (S@T-RGO)

composite, the S@B-EEG composite exhibits a high capability of 1018 mAh g⁻¹ at 1 C and excellent capacity retention of 838 mAh g⁻¹ after 130 cycles. On one hand, the excellent electrochemical performances of S@B-EEG are primarily attributed to the few defects and big planar size of B-EEG, which is helpful for increasing the conductivity of S and suppressing the shuttle of long-chain polysulfides. On the other hand, the doped boron atoms as active sites can efficiently trap polysulfides by chemical adsorption.

Keyword: Electrochemical exfoliation, boron doping, graphene, shuttle effect, lithium sulfur batteries

Introduction

After conquering the market of smart electronics, rechargeable lithium ion batteries (LIBs) are now rapidly stepping into the fields of electric vehicles (EVs) and grid-class energy storage systems. Nevertheless, the energy density of LIBs is now approaching its limits and has become one of bottlenecks that hamper the developments of batteries.¹⁻³ In this regard, lithium-sulfur (Li-S) batteries are highly attractive because of their ultrahigh theoretical specific capacity (1675 mAh g⁻¹) and energy density (2600 Wh kg⁻¹ or 2800 Wh L⁻¹).⁴⁻⁶ More attractively, S is non-toxic, low cost and natural abundant, which also makes it promising for large scale applications.⁷ Nevertheless, commercialization of Li-S batteries is still a big challenge because of some critical issues. Firstly, the inferior electronic conductivities of S (5×10⁻³⁰ S cm⁻¹) and Li₂S (10⁻³⁰ S cm⁻¹) always lead to low utilization of S and poor

rate capability.⁸ Secondly, a big volume expansion (~80%) of S cathode during lithiation/delithiation cycling causes the pulverization of electrode with fast capacity decaying.^{9, 10} Thirdly, the notorious “shuttle effect” of high-solubility long-chain lithium polysulfides (Li_2S_n , $4 \leq n \leq 8$) always results in continuous loss of S from the electrode into electrolyte, and latter diffusion to or deposition on the surface of Li metal electrode.^{8, 10, 11} This phenomenon always leads to self-discharge, rapid capacity decay and low coulombic efficiency during cycling.^{12, 13}

To overcome the aforementioned issues, graphene with high-quality has been proved to be a promising candidate due to its high conductivity, excellent mechanical flexibility, large sheet size and high specific surface area.¹⁴⁻¹⁸ Generally, high-quality graphene facilitates to encapsulate S efficiently and load S homogeneously. Meanwhile it is also beneficial for the transmission of electrons and accommodation the integrity of S upon volume expansion, which can further reduce the polarization of batteries.^{8, 19} Furthermore, the high-quality graphene can also suppress the diffusion of polysulfides into electrolyte via physical adsorption.^{20, 21} Unfortunately, recent studies have revealed that the physical adsorption based on the low binding energy between the non-polar C-C bond and polar polysulfides can't offer sufficient adsorption to the polysulfides, which further resulting in the detachment of polysulfides from the graphene matrix.^{21, 22} As a result the “shuttle effect” can not be fully suppressed and the achieved cycling stability is still unsatisfactory. To tackle this issue, surface modification via heteroatom doping (e.g. N, P, S, O and B) has been explored to anchor polysulfides more efficiently via chemical adsorption on the basement of

strong interaction between polysulfides and heteroatoms.²³⁻²⁵ For instance, B-doped graphene has been demonstrated to be helpful for trapping the soluble polysulfides via B-Li⁺ interaction.²⁵ Whereas how to synthesize high-quality heteroatom-doped graphene in order to achieve greater S utilization and enhance chemical adsorption ability still keeps a challenge.

Recently, electrochemical exfoliation method has attracted tremendous focus on production of high-quality graphene.²⁶⁻²⁸ Generally, this method is carried out in a non-aqueous or aqueous electrochemical system with one graphite electrode at least. Based on electrochemical intercalation of ions (e.g. SO₄²⁻, NO₃⁻, Li⁺, tetrabutyl ammonium (TBA⁺) etc.) between graphene layers and the subsequent expansion, this method can produce high-quality graphene (more than 60% products belong to 1~3 layers) with a large flake size (up to ~30 μm) in gram-level within hours.²⁹ More recently, the electrochemical exfoliation method has been developed to simultaneous doping graphene.³⁰ For instance, S-doped graphene is successfully prepared by electrochemical exfoliation in H₂SO₄ solution contains Na₂S₂O₃ while N-doped graphene is prepared in H₂SO₄ solution containing anthraquinone diazonium ions.^{31, 32} Inspired by these works, we herein prepare boron doped graphene sheets (B-EEG) with a B doping level of 1.86% via an electrochemical exfoliation of graphite in 1.0 mol L⁻¹ (M) Li bis(oxalato)borate (LiBOB)/dimethyl methylphosphonate (DMMP) electrolyte. The received B-EEG possesses few structure defects of I_D/I_G=0.26, large planar size of ~11 μm and low oxygen content of 7.93%. Thus, this type of B-EEG is in high quality. When the B-EEG is used as host material for S cathode, the final

S@B-EEG composite with an S content of 72.5% displays a high initial discharge capacity of 1476 mAh g⁻¹ at 0.1 C. Even at 1 C, it can deliver the discharge capacity of 1018 mAh g⁻¹ and still maintain 838 mAh g⁻¹ after 130 cycles. The excellent electrochemical performance of S@B-EEG cathode is primarily attributed to both the high-quality graphene and B-doping, which is enormously helpful for enhancing the conductivity of S, accommodating the volume expansion upon cycling and suppressing the diffusion and dissolution of polysulfides via both physical adsorption and chemical adsorption.

Experimental Section

Material Synthesis: Electrochemical exfoliation of B-EEG was performed by using an electrochemical cell with an applied voltage of 10 V. Typically, two graphite foils were used as the anode and cathode while 1.0 M LiBOB/DMMP was used as the electrolyte. After exfoliation, the electrolyte containing B-EEG was then pre-filtered and regular washed with deionized H₂O several times to remove the reserved LiBOB salt and organic DMMP solvents. Then the B-EEG was re-dispersed in N, N'-dimethylformamide (DMF) for 24h to precipitate the un-exfoliated graphite flakes. Finally, the top part of the homogeneous B-EEG dispersion was collected and further used for various characterizations and fabrication of the S@B-EEG composites.

The collected B-EEG was used to prepare S@B-EEG composite via an in situ deposition strategy. Typically, 0.075g B-EEG, 1.5 g Na₂S₂O₃ and 1 mL Trixon-100 were added in 500 mL deionized H₂O under continually magnetic stirring for 1h. Then, 70 mL HCl was slowly dropped into the solution. The as prepared S@B-EEG

composite was then pre-filtered and regular washed with deionized H₂O several times. Meanwhile, the S@thermally reduced graphene oxide (T-RGO) composites, which prepared in the same way mentioned above, is used for a comparison on electrochemical performance. Additionally, the T-RGO was synthesized from rapid heat treatment (at 1050 °C for 1 min) of the graphene oxide precursors, which was obtained in a modified Hummer's method.³³

Electrochemical Measurements: Both coin cells (2032-type) and pouch cells were used to test the electrochemical performance of the S@B-EEG composite. Pure Li foils (Φ 15.6 mm, 450 μ m thick, from China Energy Lithium Co. Ltd.) were used as the counter electrode whereas Celgard 2400 polypropylene membrane was used as separator. The cathode electrode was made by mixing 70% S@B-EEG composite, 20% ketjen black (KB) and 10% poly (vinylidene fluoride) (PVDF, Solvey HSV900) in N-methyl-2-pyrrolidone (NMP, obtained from Shanghai Aladdin Bio-Chem Technology Co. Ltd) solution, and then casted on Al foil. The electrode was then punched into disks with a diameter of 14 mm after drying in vacuum for 12 h. The S loading amount of the electrode was ~1 mg in the coin cells. For assembling a pouch cell, the mass loading of S is 15 mg with a 2.5 $\times$ 6 cm² electrode. The electrolyte was 1.0 M lithium bis(trifluoromethane sulfonimide) (LiN(SO₂CF₃)₂, LiTFSI)/1,3-dioxolane (DOL) + dimethyl ether (DME) (1:1 by volume ratio) with 2 wt% LiNO₃. To standard the test, the electrolyte amount is relative to the active mass with 35 μ L and 65 μ L per mg sulfur in coin cells and pouch cells, respectively. Galvanostatically charge and discharge measurements of the coin cell and pouch cell

were tested on a Land-CT2100A battery cycler with an electrochemical window of 1.7-2.8 V and 1.8-2.8 V, respectively. Coulombic efficiency at every cycle is defined as the charge capacity divided by the discharge capacity.³⁴ Electrochemical impedance spectra (EIS) of the batteries were performed by using a CHI604D electrochemical workstation (Shanghai Chenhua Co. Ltd.) with a frequency range from 100 kHz to 10 mHz. Cyclic voltammetry (CV) measurement with a scan rate of 0.2 mV s⁻¹ for Li-S batteries was also carried out on the CHI604D instrument. For comparison, the S@T-RGO composite was tested in the same way.

Characterization: For the visualized adsorption measurement, a solution of lithium polysulfides was firstly prepared by adding S and Li₂S (5:1 by molar ratio) in the solvents of DOL and DME (1:1 by volume ratio) under vigorous magnetic stirring at 55 °C in the glove box (MBraun, the contents of O₂ and H₂O were controlled below 0.1 ppm). Then, 10 mg B-EEG was immersed into 3 mL of the lithium polysulfides solution. Also a blank tube filled with the same amount of lithium polysulfides solution was used as the control sample. These two tubes were then placed in the glove box for 40 h, and then digital photos were taken to show the color difference of the samples.

The measurements of X-ray diffraction (XRD) were carried out on a diffractometer radiation of Bruker (Cu K_α). The morphology and energy dispersive spectroscopy (EDS) mapping were obtained by JEOL JSM-6390LA scanning electron microscopy (SEM) and JEM-2100F transmission electron microscopy (TEM). Raman was obtained on a HR Evolution instrument in the range between 3000 cm⁻¹ and 1000

cm⁻¹ by using a 10 mW helium/neon laser (532.8 nm). The N₂ adsorption/desorption apparatus (Quantachrome autosorb IQ-C) was employed to analyze the pore size distribution and specific surface area of B-EEG. Additionally, the specific surface area and the pore size distribution of B-EEG were calculated on the basement of conventional Brunauer-Emmett-Teller (BET) method and the density function theory (DFT) method, respectively. X-ray photoelectron spectroscopy (XPS) was analyzed by the instrument of America Thermo ESCALAB250. Thermogravimetric analysis (TG, Netzsch STA449F3) was carried out in a N₂ atmosphere heating from 25 °C to 800 °C at a scan rate of 5 °C min⁻¹.

Results and discussion

Heteroatom-doped high-quality graphene has been regarded as a promising host material to enhance the conductivity, reduce “shuttle effect” and accommodate the huge volume expansion of S for Li-S batteries. Nevertheless it still remains a great challenge to produce high-quality graphene with heteroatom doping to further improve the electrochemical performances.³⁵ Here, high-quality B-doped graphene is prepared by electrochemical exfoliation of graphite in 1.0 M LiBOB/DMMP electrolyte via a one-pot process and the obtained material is further employed as a host matrix for the cathode of Li-S batteries. As illustrated in **Figure 1a**, the electrochemical exfoliation method was performed in the 1.0 M LiBOB/DMMP electrolyte. Through a series of electrochemical exfoliation, about 9.52 g B-EEG was obtained within 60 min by using two graphite foils (~25 cm × 15 cm) as shown in **Figure 1b**. Therefore, the electrochemical exfoliation method can produce B-EEG in

large-scale. Morphology and microstructure of the B-EEG were further investigated by SEM and TEM. As depicted in **Figure 1c**, the pristine graphite has been efficiently exfoliated into ultrathin graphene sheets. Notably, the sheet size of the B-EEG is quite large ($\sim 11.0\ \mu\text{m}$) and the surface of the B-EEG is flat and smooth in **Figure 1c-d**. These features coincide well with the characteristics of high-quality graphene as previously reported.²⁷ Furthermore, the high-resolution TEM (HR-TEM) image in **Figure 1e** also proves the existence of the bilayer B-EEG with an interlayer spacing of $3.392\ \text{\AA}$. The overall results of SEM and TEM indicate electrochemical exfoliation of graphite in $1.0\ \text{M LiBOB/DMMP}$ electrolyte could produce high-quality B-EEG in large-scale.

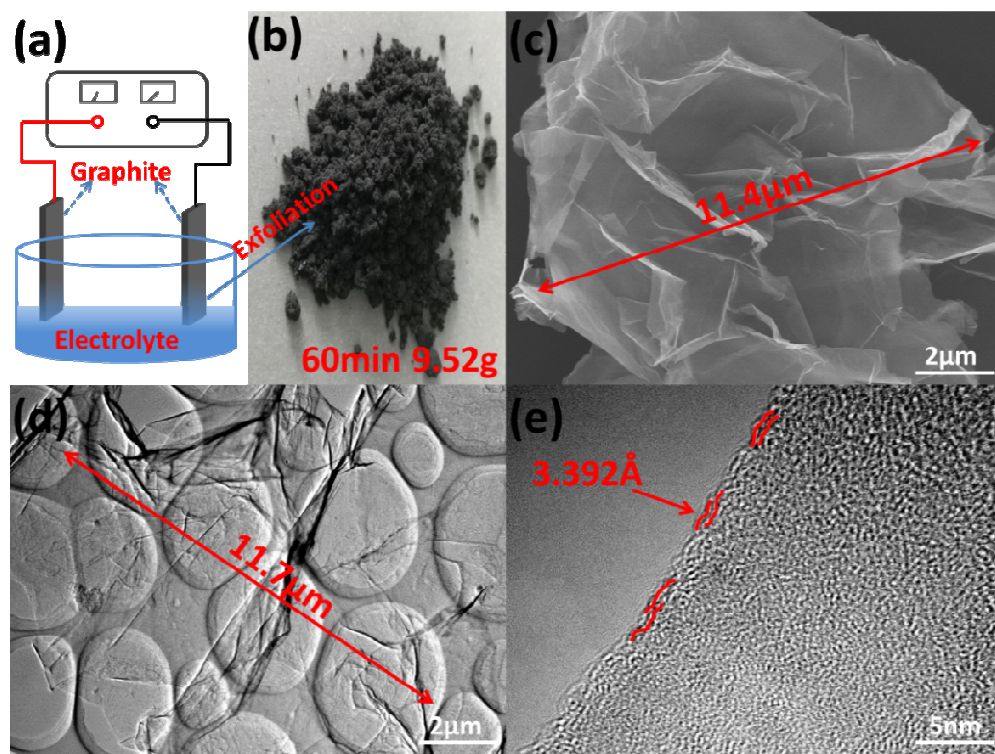


Figure 1 (a) Schematic diagram of the electrochemical exfoliation method. (b) Optical image of the dried B-EEG. (c) SEM image of the B-EEG on Si/SiO₂ substrate.

(d) Typical TEM image of the B-EEG. (e) HR-TEM at the edge of a bilayer B-EEG.

The XRD and Raman spectra were further employed to prove the high-quality characteristics of the B-EEG. As shown in **Figure 2a**, the intensity of the (002) peak decreased significantly, suggesting that the pristine graphite material has been exfoliated successfully as expected. Furthermore, the Bragg's equation ($n\lambda=2d\sin\theta$, where $n=1$, $\lambda=1.5406$ Å, d and θ is the interlayer distance and scattering angle, respectively) has also been used to calculate the spacing of the (002) planes (d_{002}).²⁷ As shown in **Figure S1a** (Supporting Information), the d_{002} layer distance of the graphene increases from 3.367 Å to 3.384 Å of the B-EEG, which coincides well with the HR-TEM results shown in **Figure 1**.

Raman was further used to reveal the structure characteristics of the B-EEG as shown in **Figure 2b**. Generally, Raman spectrum of graphene is characterized by three bands: D-band, G-band and 2D-band. Typically, the D-band is corresponding to the disordered, defect containing carbon while the G-band is attributed to the graphitic carbon.³⁶ In **Figure S1b** (Supporting Information), the wavenumber of the G-band of B-EEG (1583 cm^{-1}) is very close to that of the pristine graphite (1579 cm^{-1}), which suggests negligible oxygen content on the surface of B-EEG.³⁷ The 2D-band is tightly associated with the stacking order along c-axis and layer number of graphene.³⁶ For example, the single-layer graphene is characterized by a single and sharp 2D-band centered at 2679 cm^{-1} whereas the bilayer graphene is characterized by a broader 2D-band located at 2700 cm^{-1} . Since the 2D-band of B-EEG is observed at 2720 cm^{-1}

in **Figure S1c** (Supporting Information), we can conclude the layer number of B-EEG is probably five or more layers.³⁸ Furthermore, previous literature has reported the intensity of the D-band to the G-band (defined as the ratio of I_D/I_G) is associated with the content of structural defects while I_{2D}/I_G , the intensity ratio between 2D-band and G-band, reflects the degree of graphitization.³⁶ The higher degree of graphitization also means the fewer structural defects. The calculated I_D/I_G ratio of B-EEG is 0.26, which is much smaller than that of conventional reduced graphene oxide (RGO) (1.1~1.5) while the I_{2D}/I_G ratio (0.79) of B-EEG is also higher than that of RGO (0.3~0.75) as reported previously.^{39, 40} Both ratios can prove few structural defects of B-EEG.

XPS was further used to confirm the B doping nature of B-EEG. Since the wide scan of XPS survey can provide a full view on the chemical composition of B-EEG, the existence of B1s peak suggests that B atoms are successfully doped into graphene as shown in **Figure 2c**. Meanwhile, as shown in **Figure 2d**, the atom ratio of C, O, and B in the B-EEG is 90.21%, 7.93% and 1.86 at%, respectively. Attractively the atomic ratio of O is only a little higher than that of the raw graphite material (6.66%). Therefore, the electrochemical exfoliation method would not introduce many oxygen functional groups on the surface of B-EEG, which can further be confirmed by the high-resolution spectra of O1s in **Figure S1d** (Supporting Information). It can also be seen that the atomic ratio of C to O (C/O) is 11.37, much higher than that of RGO (3~9) reported previously.³⁹ Furthermore, the high-resolution spectra of B1s can also prove the successful doping as the two fitted peaks of $-BC_3$ (191.9 eV), $-BC_2O$ and

-BCO₂ (192.6 eV)²⁵ in **Figure 2e**. From the high-resolution spectra of C1s in **Figure 2f**, the peak located at 284.6 eV is mainly arisen from sp² carbons of graphene (C=C) whereas other peaks located at 285.4 eV and 286.5 eV are ascribed to the C-OH and C-O-C functional groups of sp³ carbons, respectively.⁴¹ In addition, the peak located at 283.9 eV of C-B group also indicates successful doping of B atoms.

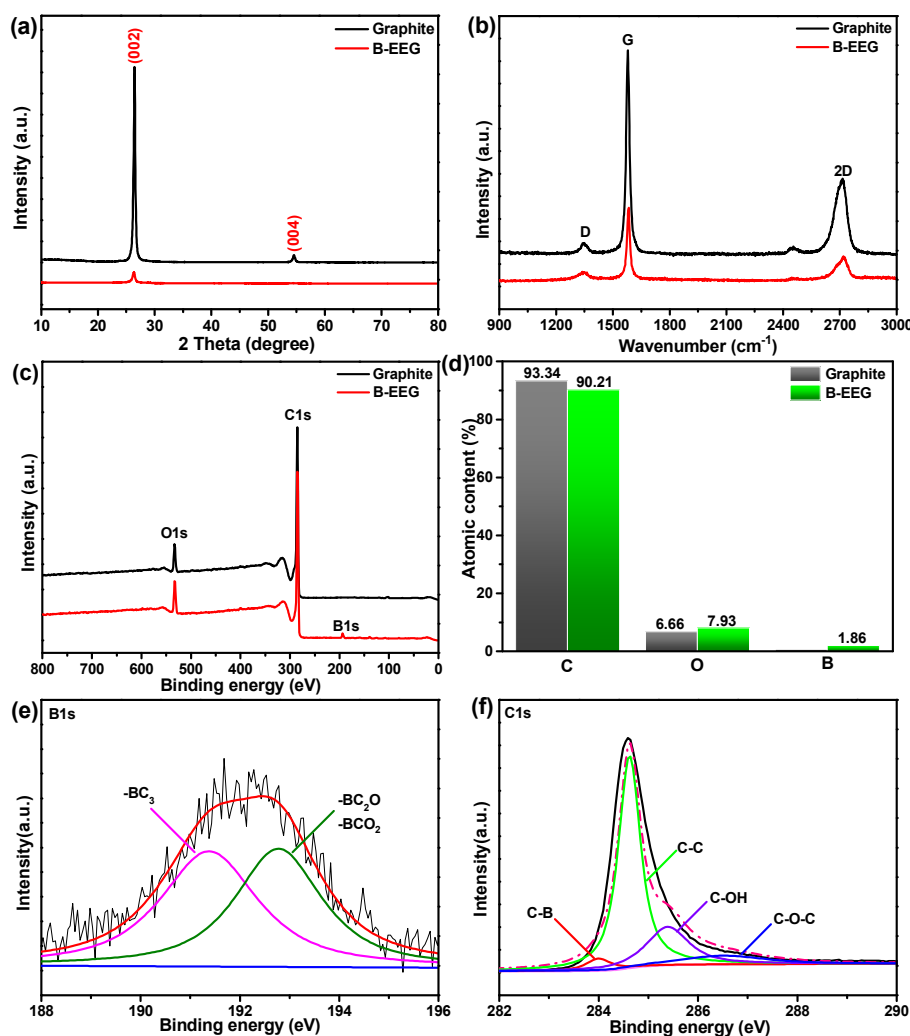


Figure 2 Structural characterization of the pristine graphite and B-EEG: (a) XRD patterns. (b) Raman spectra. (c) XPS survey scans. (d) Element contents. (e) B 1s spectra of the B-EEG. (f) C 1s spectra of the B-EEG.

N_2 adsorption-desorption isotherm is further employed to analyze the specific surface area and the pore size distribution of the B-EEG. As shown in **Figure 3a**, the specific area and pore volume of B-EEG is $70.2 \text{ m}^2 \text{ g}^{-1}$ and $0.33 \text{ cm}^3 \text{ g}^{-1}$, respectively. Notably, the specific area of B-EEG here is comparative with the electrochemically exfoliated graphene reported previously and its true specific area might be underestimated due to significant restacking of graphene sheets.^{32,42} Meanwhile, the B-EEG exhibits a typical IV N_2 adsorption/desorption isotherm with the major pore size distributions between 2.5 nm and 6 nm as shown in **Figure 3b**, which corresponds to a small mesoporous structure.

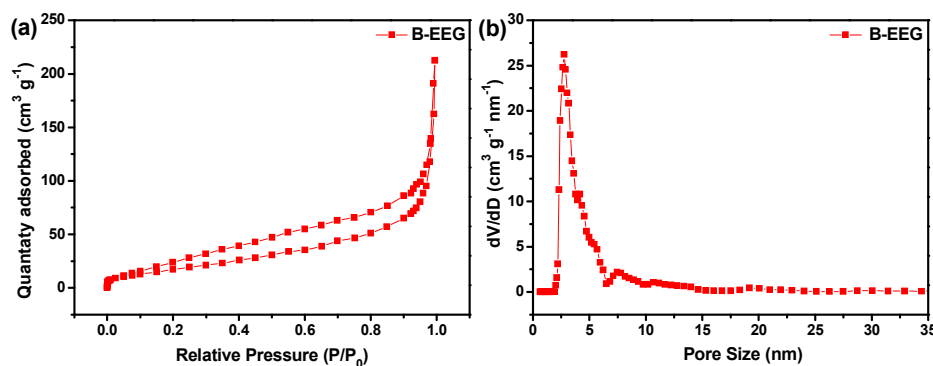


Figure 3 (a) The N_2 adsorption and desorption isotherm, (b) the pore size distribution of B-EEG.

In view of the merits of high quality and B-doping nature mentioned above, the B-EEG would be capable to enhance S utilization and anchor polysulfides through both physical adsorption and chemical adsorption for Li-S batteries with high performance. In this case, characteristics of the synthesized S@B-EEG composite were first investigated by XRD and TG in **Figure 4**. In **Figure 4a**, the diffraction

peaks of S@B-EEG composite matched well with the standard XRD patterns of orthorhombic S crystalline (JCPDS 00-008-0247), which proved that S has been successfully loaded on the surface of graphene.⁴³ To further identify the S content in the S@B-EEG composite, TG measurements was conducted under N₂ atmosphere heating from 25 °C to 800 °C with a rate of 5 °C min⁻¹. As shown in **Figure 4b**, S starts to evaporate at ~150 °C and loss the weight completely around ~250 °C. The main weight loss within the temperature range indicates the content of S is approximately 72.5%, which is quite satisfying for practical applications. Additionally, it is notable that the weight loss above 300 °C is negligible, which can also prove few oxygen-containing function groups on the surface of B-EEG discussed above.⁴⁴

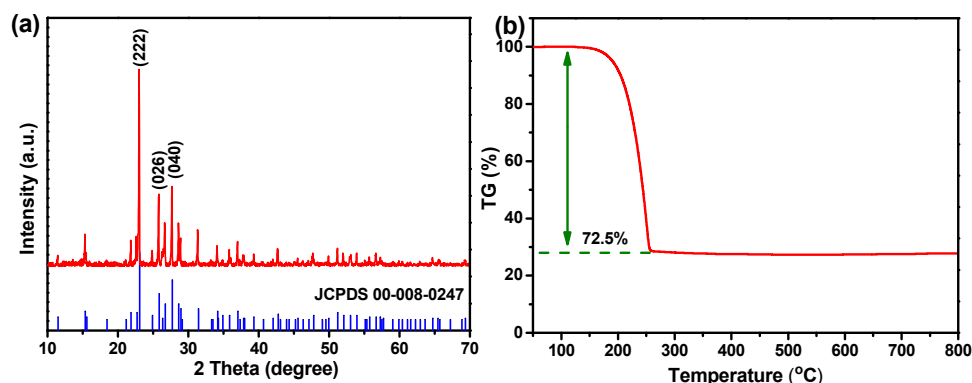


Figure 4 (a) XRD of the S@B-EEG and (b) TG curve of the S@B-EEG composite.

To identify the morphology of the S@B-EEG composite, SEM image of S@B-EEG was investigated. Benefit from the large sheet size of B-EEG, S particles are encapsulated efficiently as shown in **Figure 5a**. Meanwhile, S particles are not observed on the external surface of B-EEG. The EDS mapping in **Figure 5b-c** and the TEM image in **Figure 5d** further confirm the homogeneous distribution of S particles

on the high-quality B-EEG matrix. As a result, the B-EEG sheets can supply sufficient active sites for S and plays an important role in trapping the dissolved polysulfides as well as accommodating the volume expansion.⁴⁵ These combined merits of S@B-EEG would be capable for achieving excellent electrochemical performance as below.

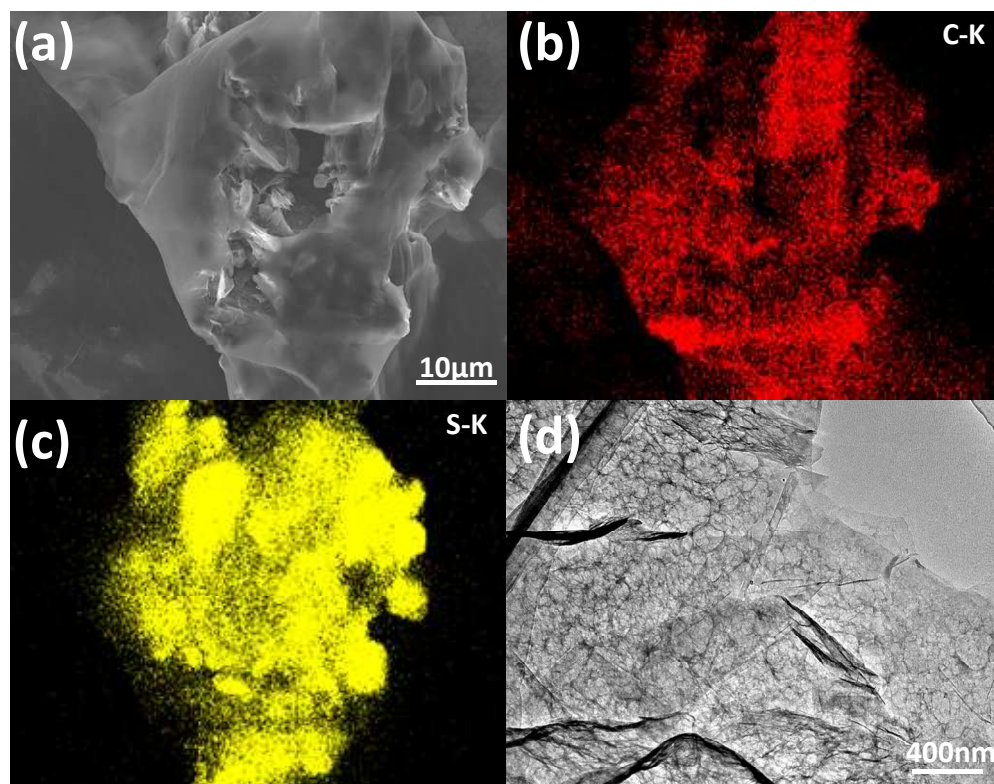


Figure 5 (a) SEM of the S@B-EEG composite; EDS mapping for element of: (b) C, and (c) S; (d) TEM of the S@B-EEG composite.

Electrochemical tests were further employed to evaluate the electrochemical performances of the S@B-EEG composite. **Figure 6a** displays the CV curves of the S@B-EEG composite. The cathodic scan shows two peaks at 2.17 and 1.95 V, corresponding to the reduction of S to soluble long-chain polysulfides (Li_2S_n , $4 \leq n \leq$

8) and then to short-chain polysulfide species (Li_2S_n , $n \leq 2$), respectively. Both the cathodic peaks show a negative shift in comparison to their theoretical values (~ 2.3 V and 2.1 V), which is caused by a large over-potential and sluggish redox kinetic. On the one hand, a large over-potential is resulted from the formation of low absorbing energy complexes between polysulfides and B-EEG conductive matrix according to Hwang's literature.⁴⁶ On the other hand, the sluggish redox kinetic is caused by the high loading of S content and the fast scan rate (**Figure S2** in Supporting Information).^{47, 48} Additionally, the small peak located at ~ 1.7 V is ascribed to the decomposition of LiNO_3 additive and the formation of solid electrolyte interface (SEI) on Li metal, which is consistent with the initial discharge plateau below 1.8 V in **Figure S3a** (Supporting Information). In the subsequent anodic scan, the peak located at 2.5 V is ascribed to the oxidation of short-chain polysulfide species (Li_2S_n , $n \leq 2$) to S. Furthermore, CV curves of the initial three cycles overlapped very well and no significant peak position is shifted, indicating the B-EEG is helpful for adsorbing soluble long-chain polysulfides and suppressing the loss of S from the electrode into electrolyte, which can in turn alleviate the "shuttle effect".⁴⁹ Therefore, the S@B-EEG is expected to have high capacity retention.

The rate performance of the S@B-EEG composite is investigated and compared with the S@T-RGO composite. As shown in **Figure 6b**, S@B-EEG can deliver reversible capacities of 1476, 1139, 918, 809, 711 mAh g^{-1} at 0.1, 0.2, 0.5, 1 and 2 C, respectively. More attractively, once the rate suddenly returned back to 0.5 C, the discharge capacity can also recover to 801 mAh g^{-1} , suggesting that the structure of

S@B-EEG electrode remains stable even at high rates. In contrast, the S@T-RGO exhibited poor rate performance. The better rate performance of S@B-EEG is attributed to the reduced electrochemical polarization and impedance by the high-quality B-EEG (Figure S3 in Supporting Information). In Figure S3b (Supporting Information), the S@B-EEG electrode has both lower ohmic impedance (the intercept at Z-real axis of the high frequency range) and charge transfer impedance (the semicircle of the EIS curve range from the high-to-medium frequency) than the S@T-RGO. The relatively low resistance indicates the improved conductivity and good Li^+ kinetic by the high-quality B-EEG.⁵⁰ These results to some extent reveal the quality of graphene is of critical important to enhance S utilization for Li-S batteries.

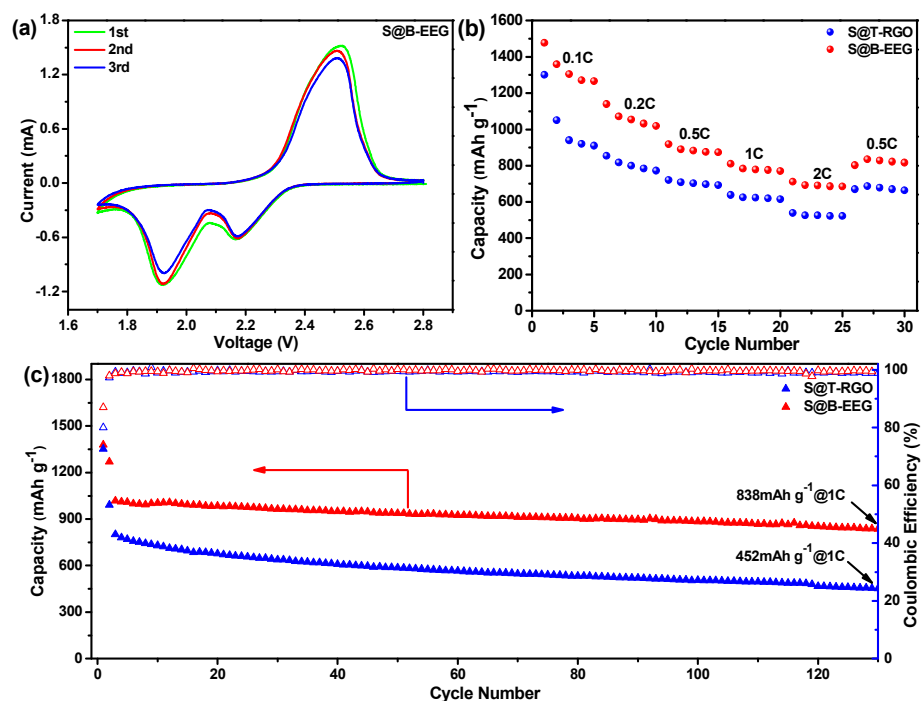


Figure 6 (a) CV curves of S@B-EEG with a scan rate of 0.02 mV S⁻¹. (b) Rate performance of S@B-EEG and S@T-RGO. (c) Cycling performance and coulombic

efficiency of S@B-EEG and S@T-RGO. The cells were cycled at 1C after two formation cycles at 0.1C.

Inspired by the good rate capability, the cycling stability of the S@B-EEG and S@T-RGO composites are further compared. To directly highlight the important role of the quality of graphene on improving the electrochemical performance of S cathode, both of the cells were directly cycled at a high rate of 1 C after two activation cycles at 0.1 C. An initial capacity of 1018 mAh g⁻¹ can be obtained at 1 C for S@B-EEG. After 130 cycles, the discharge capacity decreased to 838 mAh g⁻¹, with a decaying rate of only 1.3% per cycle in **Figure 6c**. Except the lower coulombic efficiencies at initial several cycles because of the decomposition of LiNO₃ and the SEI layer formation, the coulombic efficiency of the cell maintained about 99.5%. In contract, the S@T-RGO composite exhibits lower discharge capacity and poorer stability. After 130 cycles, its discharge capacity is only 452 mAh g⁻¹. More attractively, the SEM images of the cycled S@B-EEG electrode in **Figure S4** (Supporting Information) indicates the structural stability of S@B-EEG electrode upon repetitive cycling, which to some extent proves the positive effect of B-EEG on accommodating the volume expansion of S upon cycling.^{51, 52}

Since the long-chain polysulfides would possess more time for dissolving and diffusion at lower rates, the S@B-EEG electrode was further tested at a low rate of 0.5 C to understand the trapping ability of B-EEG toward polysulfides. In **Figure S5a** (Supporting Information), a reversible capacity of 585 mAh g⁻¹ can be obtained even

after 250 cycles. Meanwhile, the coulombic efficiency is remained at $\sim 99.3\%$ during cycling process, suggesting that the shuttle effect has been constrained to a quite low level even at low rate. Additionally, both the voltage gap of the charge-discharge curves in **Figure S5b** and low charge transfer impedance in **Figure S5c** indicate a low polarization of S@B-EEG. As a result, the B-EEG can adsorb polysulfides more efficiently, thus polysulfides dissolution can be suppressed and cycling performance of the cell improved. For commercial applications, high areal mass loading is critical for the fabrication of Li-S batteries with high energy density. The electrochemical performance of S@B-EEG was tested with a mass loading of 2.2 mg cm^{-2} (**Figure S6** in Supporting Information). Even though the S@B-EEG at the high mass loading delivers lower reversible capacity than that at the low mass loading, the good cycling stability definitely suggests that the S@B-EEG is attractive after its composition optimization. We have also compared the electrochemical performances of S@B-EEG with other S/graphene materials reported previously in **Table S1** (Supporting Information). The cell performance of S@B-EEG is also comparable with those high-performance S/graphene materials with the high S content.^{12, 49, 53-57}

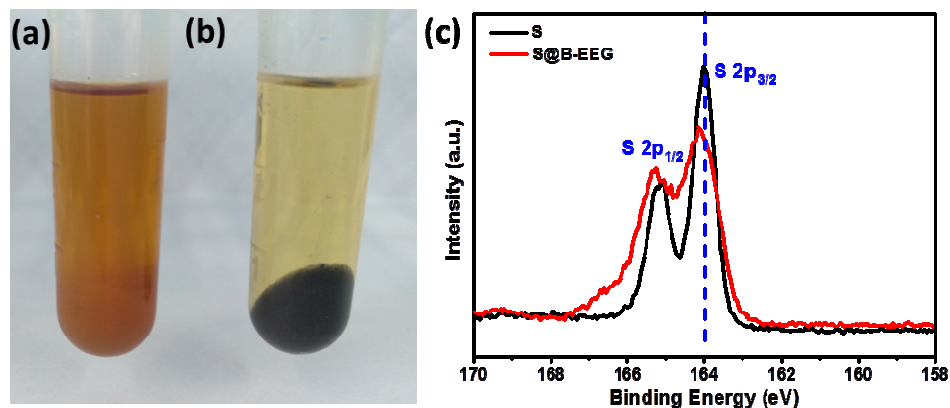


Figure 7 Optical images of the (a) blank Li_2S_n solution and (b) Li_2S_n solution

containing B-EEG after 40h. (c) S 2p XPS spectrum of the S@B-EEG electrode and pure S.

To more visually verify the trapping ability of B-EEG toward polysulfides, an adsorption test was performed. As shown in **Figure 7b**, the color of the electrolyte solution containing B-EEG is much lighter and more transparent after 40 h adsorption in comparison with that of the control sample in **Figure 7a**. Therefore, it can be concluded that the B-EEG possesses an excellent trapping ability toward polysulfides. XPS of the S@B-EEG electrode is further adopted to understand the intercalation between polysulfides and B-EEG. The S@B-EEG electrode is firstly discharged to 1.9 V to produce polysulfides according to previous investigation.⁵⁵ After discharge, the binding energy of the high-resolution S 2p spectra of the S@B-EEG is slightly higher than that of the pristine S as shown in **Figure 7c**. This suggests that the electron deficient B atoms in B-EEG accept electrons from polysulfides, leading to strong chemical interaction between polysulfides and B-EEG.^{21, 25, 55}

The above evidences clearly substantiate the advantages of B-EEG on enhancing the performances of Li-S batteries. As illustrated in **Figure 8**, during the lithiation process, Li^+ and electron transfer to the graphene and further reduce S to Li_2S , coupled with the formation of soluble long-chain polysulfides. Compared to the T-RGO, the B-EEG has several advantages, such as few layers, large sheet size, few structure defects, good conductivity and B doping. When used as a host matrix for S cathode, large-sized few-layer B-EEG facilitates to encapsulate S efficiently and thus

reduces the dissolution of long-chain polysulfides via physical adsorption. Specifically, the doped B atoms could also create more active sites, which can in turn help trap polysulfides through chemical adsorption. Meanwhile, the good conductivity can supply sufficient electrons to the anchored polysulfides so that the polysulfides can be reduced in time.⁵⁸ Additionally, the good mechanical property is also beneficial for accommodation the volume expansion/shrink upon lithiation and de-lithiation of S.¹⁹ Benefiting from these multi-function effects, the S@B-EEG exhibits improved electrochemical performances.

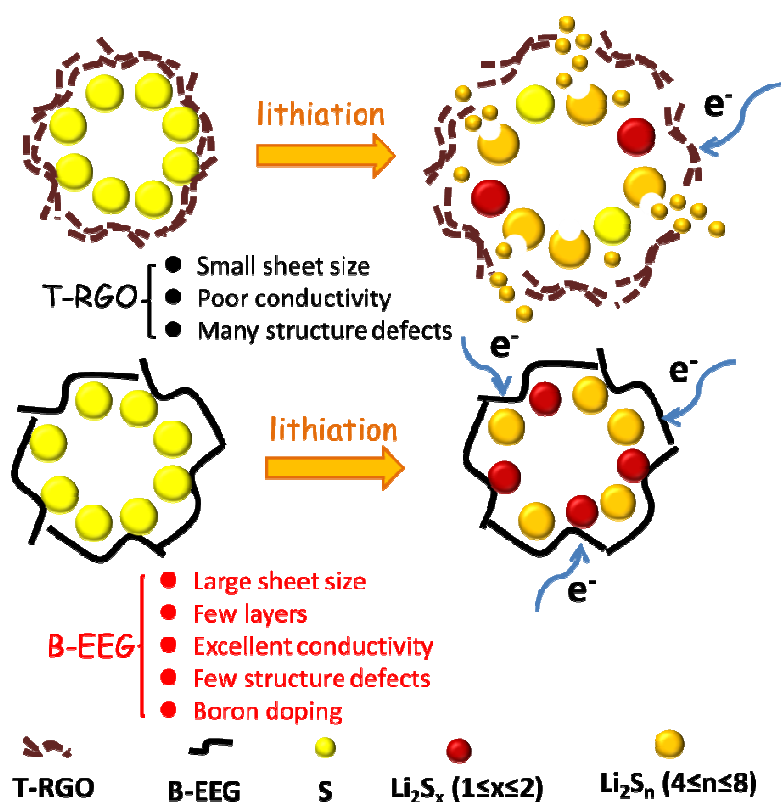


Figure 8 Schematic illustration of the advantage of B-EEG on improving the electrochemical performance of S cathode.

To further validate the potential for practical application of the S@EEG composite, Li-S pouch cell with a mass loading of ~ 15 mg was also assembled. As shown in **Figure 9a-b**, the pouch cell can output enough energy to light an LED lamp for more than 5h. Since the decomposition potential of LiNO_3 increased to ~ 1.8 V in pouch cell as depicted in **Figure 9c**, the cut off voltage was set between 1.8-2.8 V. The pouch cell shows an initial discharge capacity of 12.2 mAh and a charge capacity of 14.6 mAh. Here, the initial charge capacity is much higher than the discharge capacity because the shuttle effect has not completely suppressed since more electrolyte (corresponding to a higher ratio of electrolyte-to-S) was used in the pouch cell. After 25 cycles, the capacity decreased to 7.6 mAh with a coulombic efficiency of $\sim 98.5\%$ in **Figure 9d**.

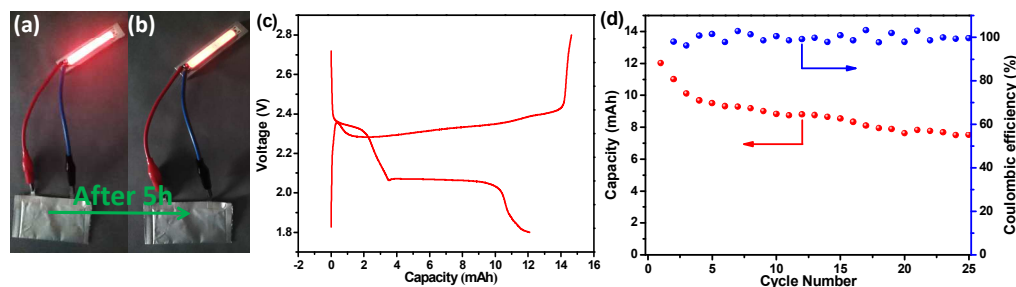


Figure 9 Electrochemical performance of the S@B-EEG in a pouch cell: (a-b) Digital photograph of cell light a LED lamp before and after 5h. (c) 1st curves of the pouch cell. (d) Cycling performance of the pouch cell.

Conclusions

In this work, electrochemical exfoliation method has been developed to simultaneous doping graphene in 1 M LiBOB/DMMP electrolyte. The as-made

B-EEG possess large sheet size ($\sim 11.0 \mu\text{m}$), low oxygen content (7.93%) and few structure defects ($I_D/I_G=0.26$), which is beneficial for enhance S utilization and capable for suppress the dissolution of polysulfides via physical adsorption. Specifically, the doped B atoms could create more active sites, which can in turn help trap polysulfides more efficiency through chemical adsorption. Benefit from these merits, the prepared S@B-EEG material with an S content of 72.5% displayed a high initial discharge capacity of 1476 mAh g^{-1} at a rate of 0.1C. When the rate increased to 1C, a high discharge capacity of 1018 mAh g^{-1} can be obtained and remain at 838 mAh g^{-1} even after 130 cycles. Moreover, a high rate capacity of 711 mAh g^{-1} is still retained at 2 C. As a result, the high-quality B-EEG is a promising host material for S cathode of Li-S batteries.

ASSOCIATED CONTENT

Supporting Information: Enlargement of the (002) peak of B-EEG in XRD; Enlargement of the G-band and the 2D-band in Raman; O 1s spectra of B-EEG and the pristine graphite; CV; electrochemical performance of S@B-EEG and S@T-RGO; SEM and the corresponding EDS images of the cycled electrode; XPS. A comparison of the electrochemical performances between this investigation and some other graphene based sulfur cathode materials reported in recent years is also included. The Supporting Information of this paper can be downloaded in the online version.

Author Information:

Corresponding author: Tel.: +86-551-62901457; E-mail: hfxiang@hfut.edu.cn

Notes:

The authors declare no competing financial interest.

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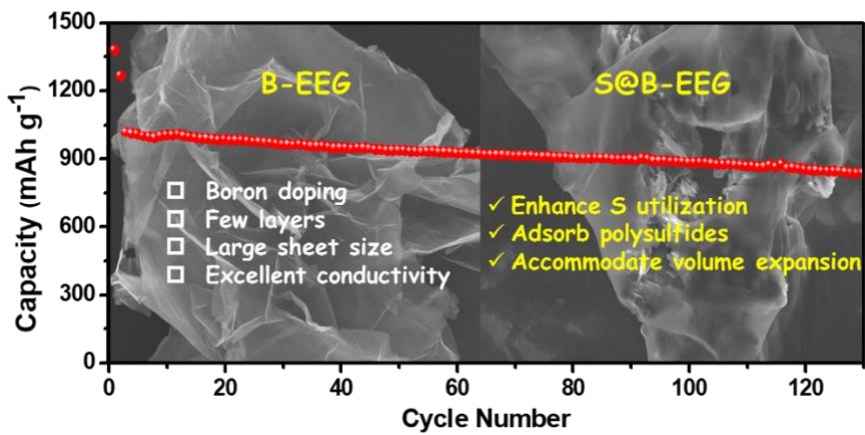
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Simultaneous boron doped graphene from electrochemical exfoliation enables high performance Li-S batteries.